

Sai Krishna Ponnamm

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EDUCATION

University of Wisconsin-Madison

Sep 2024 – May 2026

Master's in Computer Science

GPA – 4.0 / 4.0

Courses: Advanced NLP, Machine Learning, Image Processing, Big Data Systems, High Performance Computing.

National Institute of Technology (NIT), Calicut, India

July 2015 – May 2019

Bachelor of Technology in Computer Science

GPA – 8.43 / 10.0

Courses: Data Structures and Algorithms, Discrete Structures, Operating Systems, DBMS, Computer Networks, Distributed Computing, Computer Intelligence, Pattern Recognition, Graph Theory.

SKILLS

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|-------------------------|---|
| • Programming languages | C, C++, Java, Kotlin, Python, Scala, JavaScript, SQL, Bash. |
| • Databases | Redis, Cassandra, MongoDB, Hive. |
| • Search Technologies | Lucene, Elasticsearch, Solr. |
| • Deep Learning | TensorFlow, PyTorch, Keras, HuggingFace, CUDA. |
| • Other Technologies | Spark, Kafka, OpenMP, Kubernetes, Docker, Git. |
| • Soft Skills | Team Player, Agile, Proactive, Adaptable. |

PROFESSIONAL EXPERIENCE

Target Corporation, India

July 2019 – July 2024

Senior Software Engineer

Nov 2021 – July 2024

Guided Search (Facets, Autocomplete, Related Suggestions) - Search

- Architected and built a lexical-based NER system in **Python** to identify attributes within user queries, reducing manual tagging by 50%.
- Fine-tuned Llama2-70B for NER task using PyTorch and **QLoRA**, boosting attribute coverage by 12%.
- Migrated autocomplete from Trie to **Lucene FST**, reducing response time by 10ms and cutting CPU usage by 30%.
- Automated user interaction metrics pipeline for facets using **Scala Spark**, enabling data-driven roadmap planning and improving feature targeting, which reduced analysis time by 80% and accelerating planning cycles.
- Designed and developed an **A/B test** analysis framework with **Spark** and **Python** to streamline impact assessment and accelerate product decisions, reducing turnaround time by 40% and supporting 10+ experiments per year.
- Reimplemented related queries pipeline in Scala Spark, achieving 8x faster data generation (from 8 hours to 1 hour).

Software Engineer

Jun 2019 – Nov 2021

Autocomplete - Search

- Implemented a context model using LSTM with **Keras** and **Python** to personalize autocomplete suggestions, increasing attributable demand by 2% (~\$30M/year) and CTR by 10%.
- Built a real-time Kafka pipeline in **Kotlin**, with monitoring via **Grafana**, processing 150M+ user events per day, powering timely and accurate guest context generation for improved system responsiveness.
- Automated deployment of a multi-node **Redis** cluster on **Kubernetes**, cutting deployment time from 1 hour to 10 minutes.
- Ran performance profiling and optimization for search services using JMeter, reducing monthly operational costs by 15%.

ACADEMIC PROJECTS

Mini Torch framework ([GitHub](#))

- Developed in C++ with CUDA employing shared memory and reduction techniques for computational optimization.
- Implemented core neural network operations such as matrix multiplication, activation functions, and backpropagation.

Tokenizer Transfer for Multilingual Factual Knowledge Retrieval Task ([GitHub](#))

- Implemented a Vocabulary-Free Multilingual Neural Tokenizer.
- Utilized FVT and FOCUS to adapt a large tokenizer into a language-specific tokenizer.

Real-Time Object Detection using MobileNets and Deconvolution

- Developed a real-time object detection application using MobileNets architecture in TensorFlow.
- Enhanced detection accuracy by applying deconvolution, improving small object recognition in low-resolution inputs.

LEADERSHIP & ACHIEVEMENTS

- Led a cross-functional learning group since 2021, organizing bi-weekly research paper discussions to foster knowledge sharing on machine learning and search systems.
- Secured 3rd place in the Target Hackathon (Oct 2023) for developing an LLM-powered NER system for search queries.
- Awarded for end-to-end implementation and optimization of the context feature in autocomplete (2020).
- Recognized with the “Be One Team” award in 2019 for exceptional collaboration and teamwork.
- Volunteered as a Python programming mentor in the Uplift program, helping individuals from non-technical backgrounds.